

# Ian Leung

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## EDUCATION

### University of Waterloo

Expected 2028

*Bachelor of Computer Science and Finance - Double Major*

GPA: 3.8/4.0

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Statistics Minor

**Relevant Coursework:** Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Calculus, Probability, Statistics, Proofs, Financial Markets & Data Analytics

## SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, VictoriaMetrics, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS EC2, OpenRouter, Jupyter, Modal, Supabase, Vercel

## WORK EXPERIENCE

### Roblox

Jun 2026 – Present

*Software Engineer Intern (Network Reliability) | Go, Python*

*San Mateo, CA*

- Built an automated CI/CD pipeline that deploys configuration changes across **5,000+** network devices in global datacenters, cutting deployment time by **96%**
- Designed and orchestrated Temporal workflows for safe, gated deployment of configuration changes, validating changes on canary devices before progressive fleet-wide rollout with **30+** parallel workers
- Identified and fixed a resource leak in an internal network service, preventing API lockout and recovering **7%** CPU utilization across devices

### Waterloo Quant Club

Jan 2026 – Present

*Software Developer | Rust, Python, TypeScript*

*Waterloo, ON*

- Deployed [www.portal.waterlooquantclub.com](http://www.portal.waterlooquantclub.com) to AWS EC2 as a Docker Compose stack (Postgres, Flask, React) behind Caddy for automatic TLS, serving **500+** members
- Built a CI/CD pipeline that SSH-deploys on merge to main, rebuilds containers, and gates the release on a post-deploy health check
- Developed an email verification service using AWS SES, sending **100+** emails per day to new members and automating account verification

### Sunnybrook Research Institute

Jul 2025 – Aug 2025

*Software Research Intern | Python*

*Toronto, ON*

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework which generates synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Made a logging API to validate test results against baselines, catching **7** software defects before production release

## PROJECTS

**Data Science Club Website** | TypeScript, React, Supabase, HTML, CSS

- Maintained [www.uwdatascience.ca](http://www.uwdatascience.ca), UWaterloo's data science club website serving **600+** members
- Replaced the executive team's manual term-start **Google Form** with an in-app onboarding flow backed by a normalized **Supabase/Postgres** schema that autofills profile and role data from existing records

**PeerAssist - Full-Stack Peer Reviewing** | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

**Trust No Ghost - 3D Maze Exploration Game** | Unity, C#, Itch.io

- Shipped **Trust No Ghost** for **60+** users at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Harnessed Unity and C# to develop expanding maze designs, AI pathfinding algorithms, and dynamic moving mechanics